



UNIVERSITEIT VAN PRETORIA  
UNIVERSITY OF PRETORIA  
YUNIBESITHI YA PRETORIA

Denkleiers • Leading Minds • Dikgopolo tša Dihlalefi

**COS720**  
**Technical Report**  
**Computer and Information Security I**  
**Insider Threat Detection**  
**InsiderEyeShield**



19 May 2026

# Contents

|          |                                                 |          |
|----------|-------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                             | <b>3</b> |
| <b>2</b> | <b>System Overview</b>                          | <b>3</b> |
| <b>3</b> | <b>Installation Instructions</b>                | <b>3</b> |
| <b>4</b> | <b>Software Requirements and Dependencies</b>   | <b>3</b> |
| 4.1      | Data Science & Machine Learning . . . . .       | 3        |
| 4.2      | Core data tools . . . . .                       | 4        |
| 4.3      | Machine learning . . . . .                      | 4        |
| 4.4      | Visualisation . . . . .                         | 4        |
| 4.5      | Plotting libraries . . . . .                    | 4        |
| 4.6      | Image processing . . . . .                      | 4        |
| 4.7      | Web & APIs . . . . .                            | 4        |
| 4.8      | Web frameworks . . . . .                        | 4        |
| 4.9      | HTTP clients . . . . .                          | 4        |
| 4.10     | App building . . . . .                          | 4        |
| 4.11     | Jupyter & Notebooks . . . . .                   | 5        |
| 4.12     | Jupyter stack . . . . .                         | 5        |
| 4.13     | Notebook utilities . . . . .                    | 5        |
| 4.14     | Data Formats & Utilities . . . . .              | 5        |
| 4.15     | Serialisation . . . . .                         | 5        |
| 4.16     | Dev utilities . . . . .                         | 5        |
| 4.17     | HTML & scraping . . . . .                       | 5        |
| 4.18     | System & Runtime . . . . .                      | 5        |
| 4.19     | Python runtime . . . . .                        | 5        |
| 4.20     | Async & networking . . . . .                    | 5        |
| 4.21     | Deployment and Execution Instructions . . . . . | 6        |
| 4.22     | Source Code Documentation . . . . .             | 6        |
| <b>5</b> | <b>Description of Trained Model Files</b>       | <b>6</b> |
| <b>6</b> | <b>Troubleshooting Notes</b>                    | <b>7</b> |
| <b>7</b> | <b>Screenshots of Prototype Functionality</b>   | <b>8</b> |

# 1 Introduction

"An insider threat is a security risk originating from within an organization. It involves a current or former employee, contractor, or trusted business partner who misuses their authorized access—intentionally or unintentionally—to compromise sensitive data, systems, or physical facilities"

InsiderEyeShield is a web-based application that utilizes Random Forest and Logistic Regression models to detect insider threats within an organization. This AI-powered insider threat detection allows a user to investigate, analyze behavioural activity using these machine learning and anomaly techniques.

The prototype uses explainability techniques such as LIME, to improve trust and interpretation of model predictions

## 2 System Overview

InsiderEyeShield was developed using Python and Streamlit for web application. The system processes behavioural datasets, applies feature engineering, scalars and machine learning models and presents through a GUI.

The Streamlit interface contains the following tabs:

- Dashboard:  
This page displays overall statistics and threat summaries, and recent alerts
- Investigation:  
This page allows the analyst to inspect suspicious user activity and log them into profiles, as well as manually input data to assess an employee.
- Analysis:  
This page provides graphical and statistical analysis of collected data
- Live Monitoring:  
This page displays a simulation of live monitoring in real-time
- Reports:  
This page allows users to generate and display investigation reports, over a period of time.

## 3 Installation Instructions

- Install Python 3.10 or later
- clone the project repository: `git clone https://github.com/Wadalisa/insider-threat-detector.git`
- open the project folder: `cd insider-threat-detector`
- create a virtual environment: `python -m venv venv`
- activate the environment
- install dependencies: `pip install -r requirements.txt`
- run application: `streamlit run app.py`

## 4 Software Requirements and Dependencies

Devices need to have Python 3.10+ as well as a function web browser. Dependencies needed are as follows:

### 4.1 Data Science & Machine Learning

## 4.2 Core data tools

|         |        |
|---------|--------|
| numpy   | 2.2.6  |
| pandas  | 2.3.3  |
| scipy   | 1.15.3 |
| pyarrow | 24.0.0 |
| joblib  | 1.5.3  |

## 4.3 Machine learning

|                  |         |
|------------------|---------|
| scikit-learn     | 1.7.2   |
| numba            | 0.65.1  |
| imbalanced-learn | 0.14.1  |
| shap             | 0.49.1  |
| lime             | 0.2.0.1 |
| sklearn-compat   | 0.1.5   |

## 4.4 Visualisation

2

## 4.5 Plotting libraries

|            |        |
|------------|--------|
| matplotlib | 3.10.8 |
| seaborn    | 0.13.2 |
| plotly     | 6.7.0  |
| altair     | 6.1.0  |
| pydeck     | 0.9.2  |

## 4.6 Image processing

|              |           |
|--------------|-----------|
| pillow       | 12.1.1    |
| scikit-image | 0.25.2    |
| imageio      | 2.37.3    |
| tifffile     | 2025.5.10 |

## 4.7 Web & APIs

3

## 4.8 Web frameworks

|           |         |
|-----------|---------|
| fastapi   | 0.135.1 |
| flask     | 3.1.3   |
| starlette | 0.52.1  |
| uvicorn   | 0.41.0  |

## 4.9 HTTP clients

|            |        |
|------------|--------|
| httpx      | 0.28.1 |
| requests   | 2.33.1 |
| httpcore   | 1.0.9  |
| urllib3    | 2.6.3  |
| websockets | 16.0   |

## 4.10 App building

|           |        |
|-----------|--------|
| streamlit | 1.57.0 |
| pydantic  | 2.12.5 |
| click     | 8.3.1  |
| watchdog  | 6.0.0  |

## 4.11 Jupyter & Notebooks

2

## 4.12 Jupyter stack

|                |        |
|----------------|--------|
| jupyterlab     | 4.5.6  |
| jupyter-server | 2.17.0 |
| ipykernel      | 7.2.0  |
| ipython        | 8.38.0 |

## 4.13 Notebook utilities

|             |        |
|-------------|--------|
| nbformat    | 5.10.4 |
| nbclient    | 0.10.4 |
| nbconvert   | 7.17.0 |
| jupyter-lsp | 2.3.0  |

## 4.14 Data Formats & Utilities

3

## 4.15 Serialisation

|             |        |
|-------------|--------|
| pyyaml      | 6.0.3  |
| toml        | 0.10.2 |
| protobuf    | 7.34.1 |
| json5       | 0.12.1 |
| jjsonschema | 4.25.1 |

## 4.16 Dev utilities

|             |        |
|-------------|--------|
| gitpython   | 3.1.50 |
| tqdm        | 4.67.3 |
| cloudpickle | 3.1.2  |
| networkx    | 3.4.2  |
| psutil      | 7.2.2  |

## 4.17 HTML & scraping

|                |        |
|----------------|--------|
| beautifulsoup4 | 4.14.3 |
| html5lib       | 1.1    |
| bleach         | 6.3.0  |

## 4.18 System & Runtime

2

## 4.19 Python runtime

|            |            |
|------------|------------|
| python     | 3.10.20    |
| pip        | 26.0.1     |
| setuptools | 80.10.2    |
| wheel      | 0.46.3     |
| certifi    | 2026.01.04 |
| openssl    | 3.5.6      |

## 4.20 Async & networking

|                  |        |
|------------------|--------|
| anyio            | 4.12.1 |
| httptools        | 0.7.1  |
| websocket-client | 1.8.0  |
| pyzmq            | 27.1.0 |
| tornado          | 6.5.5  |

---

## 4.21 Deployment and Execution Instructions

The prototype is executed locally using Streamlit.  
To deploy and run the application:

- Open a terminal in the project directory
- Activate virtual environment
- Execute : `streamlit run app.py`

## 4.22 Source Code Documentation

The source code is organized in the following format for maintainability and readability: The application is within

| name        | Date modified    | type                | size |
|-------------|------------------|---------------------|------|
| .git        | 2026/05/19 22:36 | File folder         |      |
| application | 2026/05/16 14:52 | File folder         |      |
| data        | 2026/05/19 17:20 | File folder         |      |
| docs        | 2026/05/06 20:12 | File folder         |      |
| figures     | 2026/05/15 17:20 | File folder         |      |
| models      | 2026/05/12 04:32 | File folder         |      |
| notebooks   | 2026/05/15 16:37 | File folder         |      |
| reports     | 2026/05/06 20:12 | File folder         |      |
| seeds       | 2026/05/12 05:16 | File folder         |      |
| src         | 2026/05/06 20:12 | File folder         |      |
| LICENSE     | 2026/03/19 13:31 | File                | 2 KB |
| README.md   | 2026/05/10 12:05 | Markdown Source ... | 3 KB |

Figure 1: Folder View

the application folder with all the pages and assets used to build the application, the notebooks folder contains all the jupyter notebooks that were used to explore, preprocess, feature engineer, build models and explain the displayed output. The data folder is all the data used in the application for current and future use and updates of the application. The figures folder contains all the images of the seed 42. and seed contains all 10 seeds that were used to experiment and test the model, with figures and models.

## 5 Description of Trained Model Files

The folder for seeds contains all figures obtained when training the model and randomly generated seed, the output was noted and the best from all 10 seeds was selected.

In the Models folder, below, how the data was encoded, the feature names and scalers during preprocessing were saved for later use. How the data was split was also extracted into a model. After training and testing the best model using F1, *best\_model.pkl* and the best model using Recall, *recall\_model.pkl* along with the information about these models in json files are stored.

| Name                                                                                                     | Date modified    | Type      | Size       |
|----------------------------------------------------------------------------------------------------------|------------------|-----------|------------|
|  best_model.pkl         | 2026/05/15 01:03 | PKL File  | 113 870 KB |
|  best_model_info.json   | 2026/05/15 01:03 | JSON file | 1 KB       |
|  encoders.pkl           | 2026/05/11 20:34 | PKL File  | 3 KB       |
|  feature_names.pkl      | 2026/05/11 20:34 | PKL File  | 1 KB       |
|  recall_model.pkl       | 2026/05/15 01:03 | PKL File  | 113 872 KB |
|  recall_model_info.json | 2026/05/15 01:03 | JSON file | 1 KB       |
|  scaler.pkl             | 2026/05/11 20:34 | PKL File  | 2 KB       |
|  splits.pkl             | 2026/05/11 20:34 | PKL File  | 61 969 KB  |

Figure 2: Model Folder View

## 6 Troubleshooting Notes

- **Streamlit does not start**  
Ensure Streamlit is installed correctly.
- **Missing libraries**  
Run `pip install -r requirements.txt`
- **Dashboard not loading**  
Verify that all required files exist.
- **Graphs not displaying**  
Check Plotly and Matplotlib installations.
- **Port already in use**  
Run on another port using `--server.port 8502` .

## 7 Screenshots of Prototype Functionality



Figure 3: Dashboard

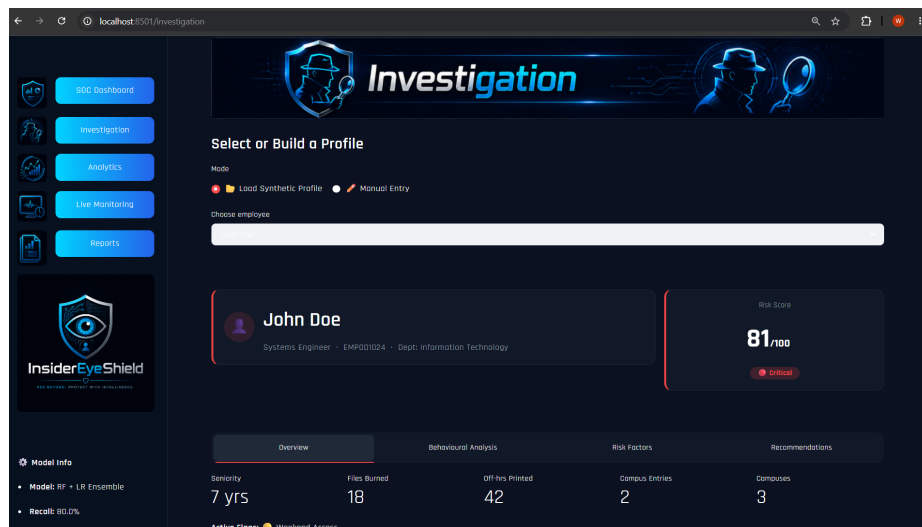


Figure 4: Profile Based Investigation

**Select or Build a Profile**

Mode: ☒ Load Synthetic Profile ☐ Manual Entry

**Employee Profile**

Department:

Campus:

Position:

Origin Country:

Seniority (years):

Privilege Classification:

☐ Is Contractor

**Behavioural Activity**

Total Pages Printed:

Pages Printed Off-Hours:

Files Burned (Removable Media):

Files Burned (From Others):

Building Entries:

Unique Campuses Visited:

☐ Weekend Building Access

**Risk Flags & Travel**

☐ Has Foreign Citizenship

☐ Has Criminal Record

☐ Has Medical History on File

☐ Currently Abroad

Trip Day Number:

Hostility Country Level:

Figure 5: Manual Entry Investigation

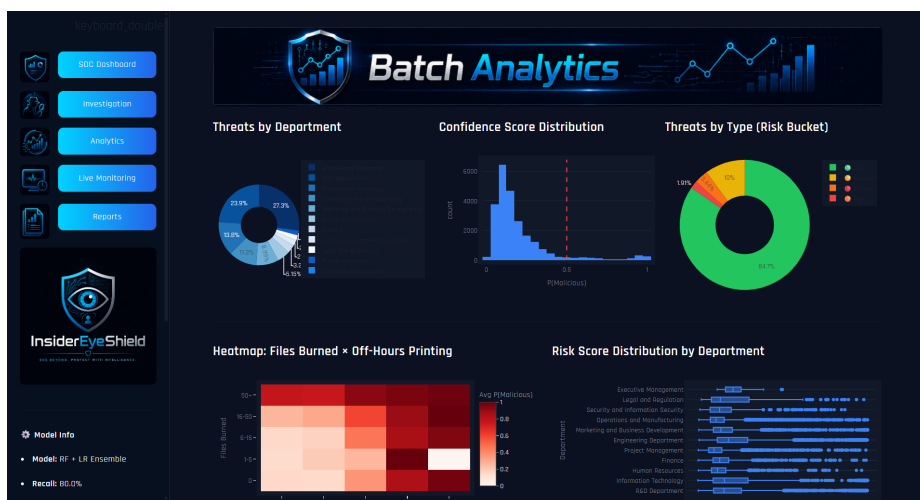


Figure 6: Analytics Page

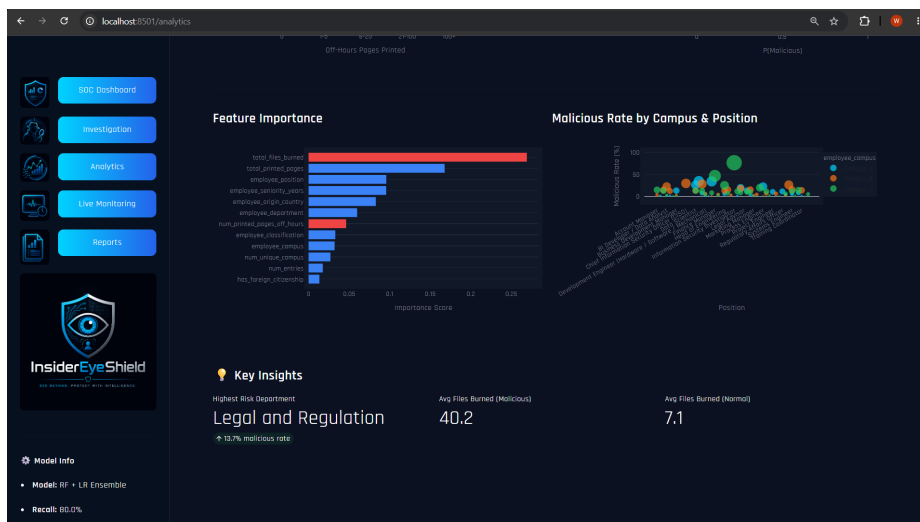


Figure 7: More Analytics Page

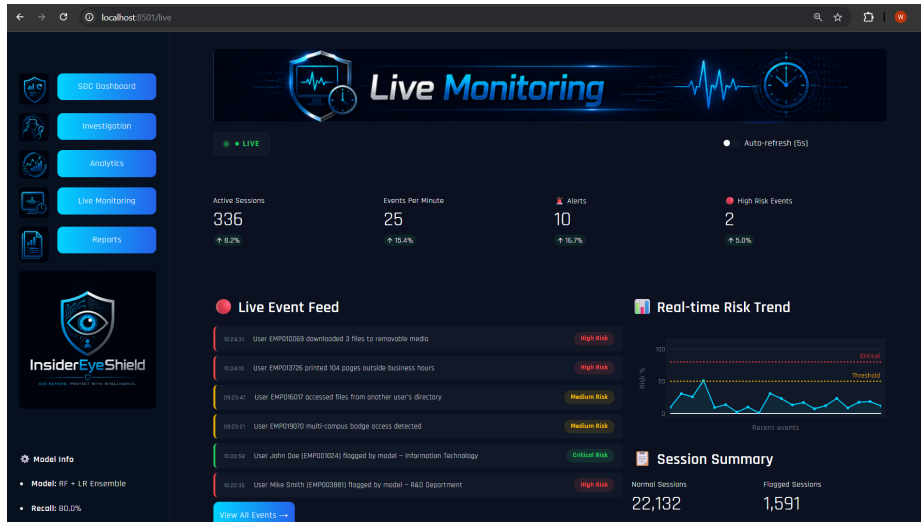


Figure 8: Live Monitoring



Figure 9: Generating Reports

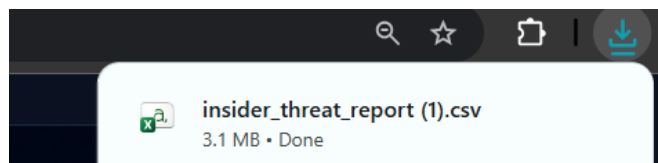


Figure 10: Downloads Reports